MUESLI: Symbolic Reasoning for Language-Based Economic Games

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Abstract

We present a symbolic-first GLEE agent that uses explicit models of incentives, uncertainty, and opponent behavior rather than an LLM at decision time. Run as the agents GRANOLA, BUCKWHEAT, and MILLET, it finished 35th of 198 agents overall, including 10th in bargaining, 35th in persuasion, and 62nd in negotiation. Bargaining performed strongly because much of its strategically relevant state is exposed through structured actions, while persuasion remained competitive using Bayesian signal reasoning and lightweight deterministic language processing. Negotiation was the clear weakness despite a richer symbolic opponent model. We argue that this contrast suggests a useful boundary: symbolic methods can go far when strategic information is structurally exposed, while language models may be most valuable for interpreting information that remains latent in dialogue.

1 Introduction

How much of a language-based economic game actually requires language? Our experience in the GLEE (Games in Language-based Economic Environments) competition suggests an answer: less than one might expect, but not none. GLEE evaluates agents in bargaining, bilateral negotiation, and repeated persuasion across 960 configurations varying horizons, information, valuations, discounting, opponent disclosure, and communication mode [10, 6]. Although natural-language communication is available throughout these games, much of the strategically relevant state is sometimes exposed directly through structured actions.

We therefore explored a deliberately *symbolic-first* approach. We call the resulting architecture MUESLI (*Model-based Utility Estimation with Strategic Lookahead and Inference*). MUESLI uses no LLM, learned policy, embedding model, or cross-game memory at decision time; instead, it reconstructs its decisions from the player-visible state and interaction history. LLMs were used during development to assist with coding and tuning, but not to select actions during competition play.

We entered it as three near-identical agents: GRANOLA and two challengers, BUCKWHEAT and MILLET, that tested narrow changes (Section 3). This comparatively simple philosophy went surprisingly far. Among the 198 agents on the final leaderboard, our submissions finished 35th overall and, most notably, 10th in bargaining. More unexpectedly, we finished 35th in persuasion, despite persuasion being perhaps the environment in which natural language appears most intrinsically important. The clear weakness was bilateral negotiation, where we finished 62nd. Table 1 summarizes the final positions. These positions rank agents only; the public leaderboard also ranks human participants, and on that combined board GRANOLA placed 42nd of 527 overall. We argue that this pattern, rather than the bargaining result alone, is the most informative outcome of the submission.

Bargaining is unusually amenable to explicit model-based reasoning: proposals reveal exact divisions, discount factors price delay, and known horizons make continuation values explicit, so the payoff-bearing actions themselves expose most of the strategic state.

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Table 1: Final leaderboard positions among the 198 agent entries (human participants excluded), taking our best agent in each scope. Lower is better.

	Overall	Bargaining	Negotiation	Persuasion
Position (of 198)	35	10	62	35

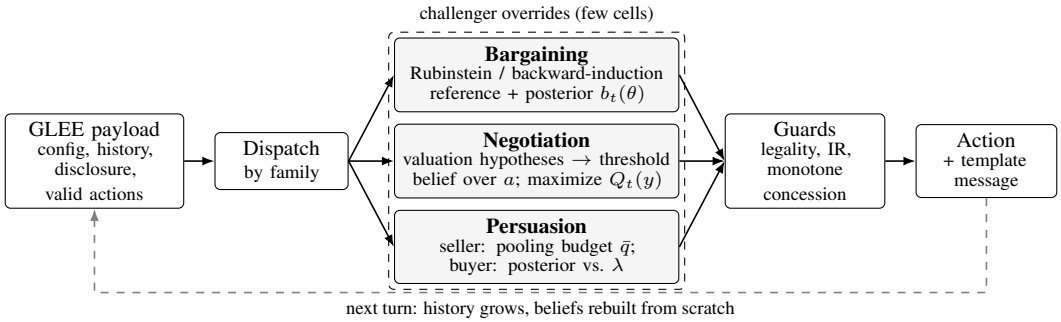


Figure 1: MUESLI decision pipeline. Each turn is a pure function of the visible payload; no LLM or cross-game memory is used.

Persuasion provides a more surprising example. Our agent does not construct arguments or use an LLM to reason about rhetoric. Instead, it treats persuasion primarily as an information-design problem: the seller controls the information conveyed by its recommendation, while the buyer reasons about the posterior probability of product quality. In effect, it persuades through information structure rather than open-ended language generation.

Negotiation exposes the clearest boundary of the approach. Private valuations leave more of the strategic state latent, while explanations, commitments, conditional proposals, bluffing, and other semantically meaningful utterances may provide evidence that cannot be recovered from prices and accept–reject decisions alone. Our negotiation policy discards almost all of this information. Its weaker result therefore suggests a natural role for language models, but not necessarily as replacements for explicit planning. An LLM could instead extract evidence from dialogue or generate strategically appropriate utterances while a symbolic core continues to reason about utilities, horizons, and feasible actions, similar to existing approaches from classical planning [7, 2].

We describe the symbolic mechanisms used by the agents and use their differing competition performance to identify the apparent boundary of this design philosophy.

2 Agent design

Pipeline. Figure 1 shows the decision pipeline shared by all three agents. Each turn, the GLEE SDK delivers the game configuration, full public history, any opponent-type disclosure, and the valid actions. The agent is stateless: it rebuilds its beliefs from this history on every call, so every decision can be replayed exactly from the logged payload. A dispatcher routes the payload to one of three family modules, each of which maps the visible state to a small explicit belief, scores candidate actions by expected utility, and passes the choice through hard legality and individual-rationality guards before an optional deterministic message is attached.

2.1 Bargaining: belief-state lookahead

Game. Two players alternate proposals for dividing a fixed sum M; the receiver accepts, rejects (passing the proposal to them), or walks away, and each player's share is multiplied by its own discount factor δ every round. GLEE varies whether the horizon is known, whether both discount factors are visible, and whether messages are allowed. Because offers and rejections are fully structured, we treat bargaining as Bayesian inference over the opponent's acceptance threshold anchored to the game-theoretic solution, rather than as a dialogue problem.

Let M be the amount to divide and let $x \in [0, 1]$ denote the share offered by proposer i to receiver j. The receiver is modelled by a latent acceptance threshold θ: an offer is expected to be accepted when $x \geq \theta$. The policy maintains a discrete posterior $b_t(\theta),\ \theta \in \{0, 0.01, \ldots, 1\}$. This deliberately small model makes opponent adaptation inspectable rather than hiding acceptance behavior inside a learned predictor.

Rational and behavioral reference points. When both discount factors are visible and the horizon is unknown, the agent uses the infinite-horizon Rubinstein receiver share as a rational reference [9],

$$x_j^R = \frac{\delta_j(1 - \delta_i)}{1 - \delta_i\delta_j}.$$

For a known finite horizon, the reference is instead obtained by backward induction. The terminal receiver has continuation value zero; working backwards, rejecting makes the current receiver the next proposer, giving the recursion $x_{j,t}^R = \delta_j \left(1 - x_{i,t+1}^R\right)$.

GLEE opponents need not follow equilibrium strategies and may instead exhibit fairness preferences, bounded rationality, or heuristic behavior [5]. We therefore combine the rational reference with a behavioral threshold prior, varying both the prior and the weight assigned to rational behavior according to whether the opponent is disclosed as a human, an agent, or remains hidden.

Learning from revealed actions. A rejection of an offer r is evidence that $\theta > r$, but we model this noisily using a tremble probability: $P(\text{reject } r \mid \theta) = 0.92$ when $\theta > r$ and 0.08 otherwise. This prevents a single anomalous action from eliminating otherwise plausible hypotheses. A rejection may also be *contradicted*: an opponent can reject a share r and later propose a division giving itself less than r. We treat such cases as weaker evidence of a reservation threshold and more plausibly as strategic delay. Likewise, an opponent's own demand is treated only as soft aspiration evidence, reflecting the distinction between preferences and bidding behavior in automated negotiation [1].

Choosing an offer. For candidate receiver share x, let $A_t(x) = P(\theta \leq x \mid h_t)$ be the posterior probability of acceptance. The agent chooses $x_t^* \in \arg\max_x \left[A_t(x)(1 - x) + (1 - A_t(x))C_t\right]$, where C_t approximates the proposer's continuation value following rejection.

The continuation model predicts the opponent's next offer from its observed proposals, falling back to the behavioral and rational models when evidence is sparse. For known horizons, the agent uses bounded lookahead to compare agreement with the value of rejecting and continuing the alternating-offer game; unknown horizons use a one-counteroffer approximation.

Acceptance uses the same continuation model: an offer is accepted when its value exceeds the discounted value of continuing, while every strictly positive terminal offer is accepted. Explicit bounds keep adaptation conservative in response to rejection evidence.

Incoming natural language does not affect the bargaining belief state. When messages are enabled, the agent instead generates deterministic, state-conditioned text explaining the selected offer using factors such as discounting, deadline pressure, and previous rejections.

Worked example. Table 2 replays a live GRANOLA game against a disclosed agent opponent with $M = \$10{,}000$, complete information, an unknown horizon, no messages, and $\delta_1 = 0.95$ (GRANOLA) versus $\delta_2 = 0.8$. The Rubinstein reference concedes only $x_j^R = 0.8 \cdot 0.05/(1 - 0.76) \approx 16.7\%$ to the more impatient opponent; blending in the behavioral prior for a disclosed agent, GRANOLA opens at 21%. The opponent's rapid concessions (52.5% then 65% for GRANOLA) shift the posterior over θ downward and raise the forecast of its next proposal, so GRANOLA rejects 65% in round 4 and concedes only one point per proposal until its 77% offer is accepted.

2.2 Negotiation: expected utility under latent values

Game. A seller and buyer alternate price proposals for one item; each has a reservation value (the seller's cost and the buyer's valuation), which may be private, and either side may accept, counter, or walk away. Unlike bargaining, the surplus itself is uncertain and may be zero, so the central problem is inferring what is feasible. We therefore combine a classical concession schedule with a belief over latent valuations, enumerated exactly where GLEE's discrete valuation grid allows it.

Table 2: Play-by-play of a live GRANOLA bargaining game (shares of \$10,000 for GRANOLA / opponent).

Round	Proposer	Split	Response and rationale
1	GRANOLA	79 / 21	Rejected; between Rubinstein (83/17) and behavioral prior
2	Opponent	52.5 / 47.5	GRANOLA rejects; far below continuation value
3	GRANOLA	78 / 22	Rejected; one-point concession
4	Opponent	65 / 35	GRANOLA rejects; \$5,573 now $<$ estimated value of countering
5	GRANOLA	77 / 23	Accepted; discounted payoffs \$6,272 / \$942

We map both roles into a common own-utility space, with $u_i(p) = p - v_i$ for a seller and $u_i(p) = v_i - p$ for a buyer, so individual rationality simply requires $u_i(p) \geq 0$.

The policy combines time-dependent concession [4] with explicit opponent modelling [1]. Let $\tau_t \in [0, 1]$ denote progress through the interaction, using normalized round number for known horizons and an exponential approximation otherwise. Under complete information with gains from trade S, we use a Boulware-style reference demand $d_t = S\left(0.50 + 0.22(1 - \tau_t^3)\right)$, beginning near 72% of the surplus and conceding more rapidly near the deadline.

Latent concession thresholds. We model a threshold a such that the opponent is expected to accept offers giving us utility at most a. Under complete information, we maintain a discrete belief over a around the time-dependent concession prior. Under private information, GLEE's discrete valuation scheme allows our own value to imply a finite set of possible opponent valuations; each induces a feasible surplus and corresponding acceptance threshold, while hypotheses with no mutually beneficial trade receive no acceptance probability.

Observed actions update this belief. An opponent offer giving us utility y supports hypotheses capable of conceding at least y and raises their estimated threshold accordingly. Conversely, rejection of our demand y provides censored evidence that the opponent's threshold lies below y. Both updates are deliberately noisy, and recent opponent concessions are also used to forecast future movement.

Expected-payoff search. For candidate own utility y, let $P_t(y) = P(a \geq y \mid h_t)$ be its acceptance probability. We select offers by maximizing $Q_t(y) = P_t(y)y + (1 - P_t(y))C_t(y)$, where $C_t(y)$ is the discounted one-step value after rejection. Candidates include a utility grid, the concession reference, observed opponent offers, and belief breakpoints. The policy also prevents concession reversal.

Incoming offers are evaluated against the certainty-equivalent value of continuing, rather than simply against the next intended counteroffer. The policy accepts sufficiently strong profitable offers, becomes more conservative about continuation after repeated stagnant behavior, and walks away when known valuations rule out positive-surplus trade or an uncapped negotiation remains persistently individually irrational.

A small number of competition-specific guards supplement this general policy, including a symmetric-surplus bracket in uncapped complete-information games and terminal behavior that exploits the disappearance of continuation value. Free-text messages are deterministic renderings of the selected price: the agent does not interpret the opponent's explanations, commitments, or tone. This omission becomes important when interpreting the negotiation result.

2.3 Persuasion: information design without an LLM

Game. Over repeated rounds, a seller observes whether each product is high or low quality and sends the buyer a recommendation, either binary or free text, before the buyer decides whether to buy at a posted price; the seller profits from every sale, while the buyer profits only from good purchases. Persuasion appears the most language-centric environment, but it also admits a compact sender–receiver interpretation [8, 3]. Let quality be high, H, with public probability p, and low, L, otherwise; the buyer values these outcomes at $v > u$ and pays price c. The model combines a symbolic signalling policy with a small deterministic text layer.

Seller policy. The model always recommends high-quality products and may pool low-quality products into a positive recommendation with probability q_t. For $u < c < v$, a positive signal induces

posterior

$$P(H \mid +) = \frac{p}{p + (1-p)q_t},$$

so buying remains profitable provided $q_t < \bar{q} = \frac{p(v-c)}{(1-p)(c-u)}$. The agent treats $\bar{q}$ as a *reputation budget*: it initially preserves informative separation and spends more of this budget later, adapting to whether the buyer responds to recommendations and to revealed low-quality purchases.

In binary games this is literal pooling, with high quality and selected low quality producing the same positive signal. In free-text games, we instead use fixed templates for high quality, strategically recommended low quality, and passing. Thus lightweight generated language provides an additional signal without requiring an LLM.

Buyer policy. The agent maps incoming messages deterministically to $\sigma(m) \in \{+, -, ?\}$, representing positive, negative, and ambiguous signals. For intermediate prices, buying requires the estimated posterior probability of high quality to exceed $\lambda = \frac{c-u}{v-u}$. Negative signals are rejected and ambiguous signals fall back to the public prior. For positive signals, we estimate $P(H \mid +)$ from the observed frequency of positive recommendations and calibrate this estimate using qualities revealed after purchases. Because quality is observed only following purchase, these outcomes are treated as selected evidence rather than as a complete sample.

The buyer purchases when the resulting posterior exceeds λ, with a small bounded exploration allowance early in long interactions. At exact public-prior indifference, the agent is deliberately conservative and requires additional evidence that positive recommendations are informative.

3 Discussion

Three agents. GRANOLA ran the policy above; two challengers tested narrow changes against it. BUCKWHEAT, preregistered before its first live game, changed three cells (accepting exactly repeated bargaining offers under discounting, a 55%/45% surplus bracket in uncapped complete-information negotiation, and extra persuasion template coverage) and otherwise delegated to GRANOLA. MILLET isolated one negotiation change, accepting profitable offers more readily after opponent evidence in capped private-value games (continuation multiplier 0.70 rather than 0.85). Modules that held up were promoted into GRANOLA, so the three agents differ only marginally.

Development process. Development mainly improved how uncertainty was represented: bargaining moved from hard reservation bounds to distinguishing reservation, aspiration, and delay; negotiation from pure concession schedules to latent valuations.

Results. The results suggest that the usefulness of language depends less on whether communication is present than on how much strategically relevant information structured actions already expose. Bargaining and, substantially, persuasion reduce to explicit allocations, signals, and beliefs; in negotiation the same price may reflect different constraints, intentions, or strategies, which motivates the hybrid design sketched in the introduction.

Agent Behavior Analysis. The policies enforce simple behavioral invariants: bargaining responds to discounting and observed actions without overcommitting to contradictory evidence; negotiation preserves individual rationality and monotone concession; and persuasion bounds strategic pooling and treats ambiguous language conservatively. Because decisions are replayable from the visible history, these invariants are directly testable; Table 2 shows the bargaining ones on a live game.

Limitations and reproducibility. Leaderboard results are observational and cannot isolate the effect of language. Code for all three agents and aggregate evidence is at `https://github.com/HarrisonOates/muesli-glee`; transcripts are withheld as they contain other participants' messages.

References

[1] Tim Baarslag, Mark J. C. Hendrikx, Koen V. Hindriks, and Catholijn M. Jonker. Learning about the opponent in automated bilateral negotiation: A comprehensive survey of opponent

modeling techniques. *Autonomous Agents and Multi-Agent Systems*, 30(5):849–898, 2016. doi: 10.1007/s10458-015-9309-1.

[2] Augusto B. Corrêa, André Grahl Pereira, and Jendrik Seipp. Classical Planning with LLM-Generated Heuristics: Challenging the State of the Art with Python Code. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, October 2025.

[3] Vincent P. Crawford and Joel Sobel. Strategic information transmission. *Econometrica*, 50(6): 1431–1451, 1982. doi: 10.2307/1913390.

[4] Peyman Faratin, Carles Sierra, and Nick R. Jennings. Negotiation decision functions for autonomous agents. *Robotics and Autonomous Systems*, 24(3–4):159–182, 1998. doi: 10.1016/S0921-8890(98)00029-3.

[5] Ernst Fehr and Klaus M. Schmidt. A theory of fairness, competition, and cooperation. *The Quarterly Journal of Economics*, 114(3):817–868, 1999. doi: 10.1162/003355399556151.

[6] GLEE Competition Organizers. GLEE Competition: Complete developer documentation. Online documentation, 2026. URL `https://glee-competition.com/llms.txt`. Accessed 2026-08-23.

[7] Subbarao Kambhampati, Karthik Valmeekam, Lin Guan, Mudit Verma, Kaya Stechly, Siddhant Bhambri, Lucas Paul Saldyt, and Anil B. Murthy. Position: LLMs Can't Plan, But Can Help Planning in LLM-Modulo Frameworks. In *Proceedings of the 41st International Conference on Machine Learning*, pages 22895–22907. PMLR, July 2024.

[8] Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6):2590–2615, 2011. doi: 10.1257/aer.101.6.2590.

[9] Ariel Rubinstein. Perfect equilibrium in a bargaining model. *Econometrica*, 50(1):97–109, 1982. doi: 10.2307/1912531.

[10] Eilam Shapira, Omer Madmon, Itamar Reinman, Samuel Joseph Amouyal, Roi Reichart, and Moshe Tennenholtz. GLEE: A unified framework and benchmark for language-based economic environments. *arXiv preprint arXiv:2410.05254*, 2024. URL `https://arxiv.org/abs/2410.05254`.